

Weighted Adjustment Instrumental Variable Estimator

Correcting for p-hacking in meta-analysis of observational studies

T. Havranek^{1,2} Z. Irsova¹ P. Bom³ H. Rachinger⁴

¹Charles University, Prague

²Centre for Economic Policy Research, London

³University of Deusto, Bilbao

⁴University of the Balearic Islands, Palma

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Changing controls changes precision

Table: Regression of test scores on class size

	(1)	(2)	(3)	(4)
Small class (treatment)	4.82 (2.19)	5.37 (1.26)	5.36 (1.21)	5.37 (1.19)
White/Asian			8.35 (1.35)	8.44 (1.36)
Girl			4.48 (0.63)	4.39 (0.63)
Free lunch			-13.15 (0.77)	-13.07 (0.77)
White teacher				-0.57 (2.1)
Teacher experience				0.26 (0.10)
Master's degree				-0.51 (1.06)
School intercepts	No	Yes	Yes	Yes
Sample	5,861	5,861	5,861	5,861

Notes: Adapted from Krueger (1999). Dependent variable: test score percentile.

Standard errors in parentheses.

Changing clustering changes precision

Table: Regression of test scores on class size

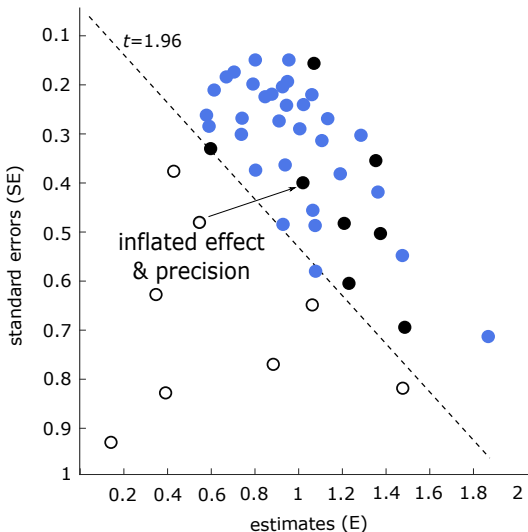
	Krueger (1999)	Replications using different computations of SE				
	(1)	(2) Bootstrap of class clusters	(3) Class clusters	(4) School clusters	(5) Huber-White SE	(6) Plain vanilla SE
Small class	4.82 (2.19)	4.71 (2.00)	4.71 (1.88)	4.71 (1.38)	4.71 (0.79)	4.71 (0.76)
Sample	5,861	5,743	5,743	5,743	5,743	5,743

Notes: Dependent variable: test score percentile. Standard errors (SE) in parentheses.

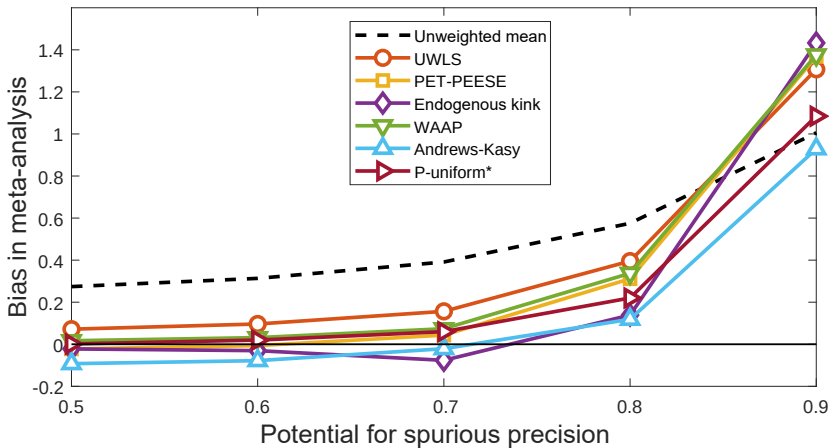
Meta-analysis of the class size effect

Available at meta-analysis.cz/class. Spoiler: effect size tiny.

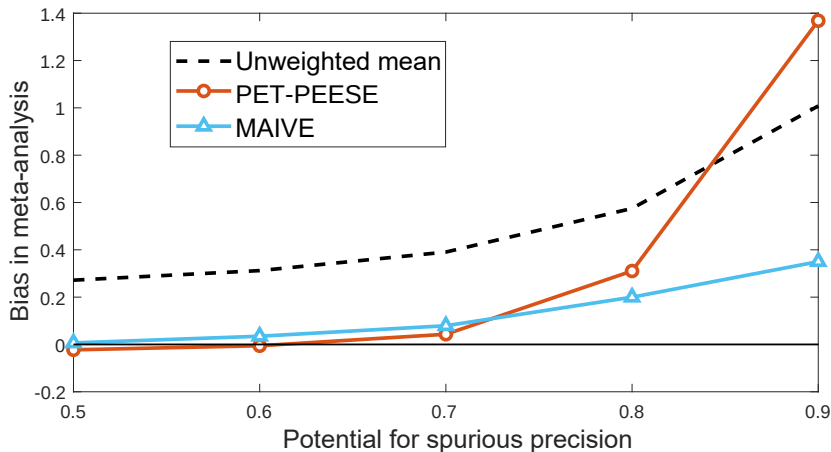
What if both types of p-hacking interact?



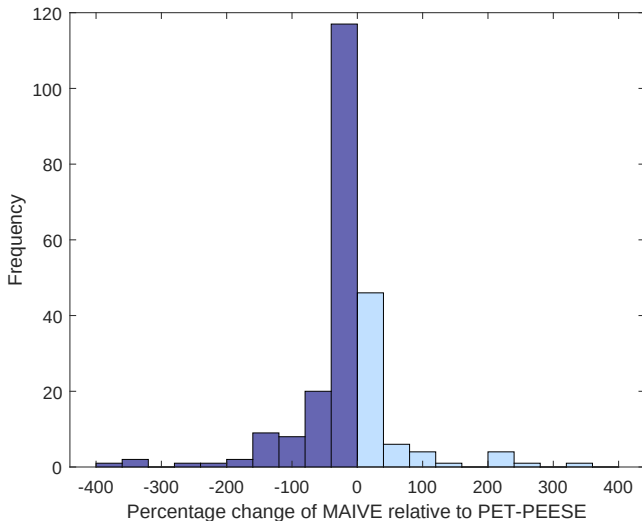
All estimators biased upwards



MAIVE alleviates the bias



MAIVE reduces PET-PEESE in 70% of the cases



Meta-analysis instrumental variable estimator

MAIVE intuition:

Definition of SE

$$\hat{E}_i = \textcircled{E_0} + \beta SE(\hat{E})_i^2 + u_i. \quad 1/N_i$$

- Can **add controls**.
- Plug MAIVE-adjusted SEs into other estimators.

Meta-analysis instrumental variable estimator

MAIVE first stage:

$$SE(\hat{E})_i^2 = \alpha_0 + \alpha_1 (1/N_i) + \pi_i.$$

hacking or misspecifications



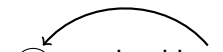
- Can **add controls**.
- Plug MAIVE-adjusted SEs into other estimators.

Meta-analysis instrumental variable estimator

MAIVE adjustment:

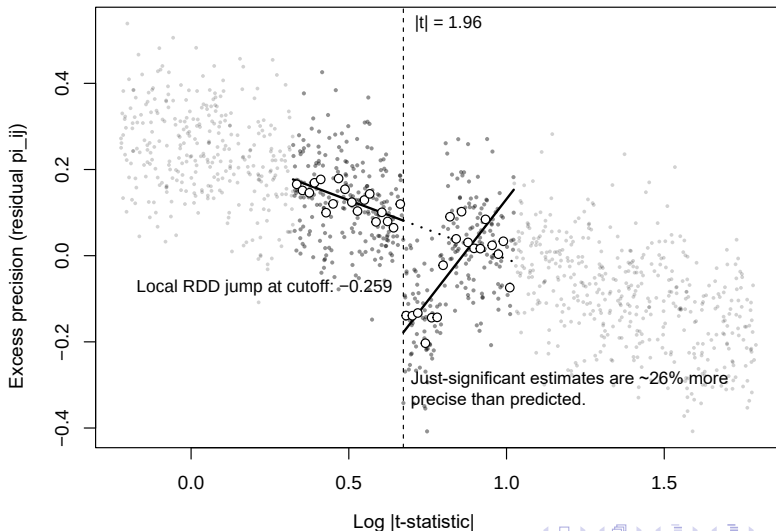
$$SE(\hat{E})_{adj,i}^2 = \hat{\alpha}_0 + \hat{\alpha}_1 (1/N_i) + \cancel{\pi_i}.$$

hacking or misspecifications



- Can **add controls**.
- Plug MAIVE-adjusted SEs into other estimators.

Residual discontinuity test



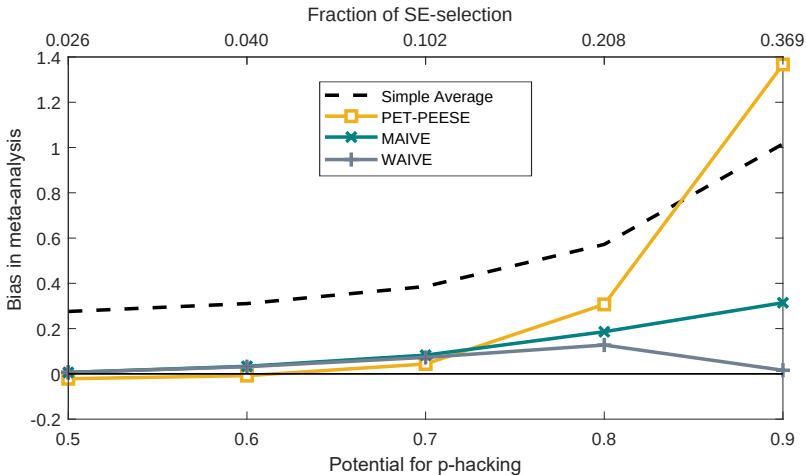
WAIVE weight for the second stage

Exponential tilt:

$$\omega_i = \exp[-\max(0, -\pi_i)]$$

- **Penalize only negative residuals** ($\pi_i < 0$) – downweight spuriously precise estimates.
- Weight $\in (0, 1]$, smooth and proportional decay.
- Then run PEESE of $\hat{E}_i$ on $SE(\hat{E})_{adj,i}^2$ with weights ω_i .

WAIVE alleviates the bias



Practical issues

WAIVE still just a concept. Also note RTMA.

Main consideration

- Can methods or p -hacking influence both $\hat{E}$ and SE ?
- Yes $\rightarrow$ MAIVE and WAIVE help (EasyMeta.org).
- No $\rightarrow$ MAIVE and WAIVE don't hurt much (but wider CI).

Project Website

meta-analysis.cz/maive

Papers using MAIVE

-  Irsova Z., P. Bom, T. Havranek, & H. Rachinger (2025):
Spurious Precision in Meta-Analysis of Observational
Research.
Nature Communications 16: 8454.
-  Irsova Z., T. Havranek, X. Bortnikova, & F. Bartos (2026):
Meta-Analysis of Field Studies on Beauty and Professional
Success.
Nature Human Behaviour, R&R.
-  Cala P., T. Havranek, Z. Irsova, M. Luskova, J. Matousek
& J. Novak (2026): Financial Incentives and Performance:
A Meta-Analysis of Experiments in Economics.
Journal of Political Economy Microeconomics, forthcoming.